

Generation Interconnection System Impact Study Report

For Queue Project Q16-0707

Hills 100 kV
October 24, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0707. The Interconnection Customer proposes a 212.4 MW solar photovoltaic facility interconnecting at the Hills 100 kV substation in the Ohio area. The request was received on November 25, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 241 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,685,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Twinsburg / Chamberlin 100kV Line 1	Chamberlin / Twinsburg 100kV L	142.1	159	8.3
Chamberlin / Twinsburg 100kV Ckt 2	Chamberlin / Twinsburg 100kV Circuit 1	142.1	159	8.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Twinsburg / Chamberlin 100kV Cir. 1	Rebuild line with higher-capacity conductor	\$1,270,000	78% (\$995,000)
Chamberlin / Twinsburg 100kV Ckt 2	Rebuild line with higher-capacity conductor	\$2,159,000	78% (\$1,690,000)
	Total allocated to project		\$2,685,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.